Research Statement

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Research Vision

My research develops **trustworthy, secure, and data-efficient artificial intelligence** for high-stakes medical and biomedical decision support. I work at the intersection of **machine learning**, **hybrid quantum-classical computing**, **medical image analysis**, **secure/adversarial machine learning**, and **AI systems**. The central goal of my research is to design learning systems that remain reliable when real-world data are imperfect, including noisy labels, corrupted imaging inputs, weak or incomplete supervision, distribution shift, and training-time backdoor poisoning.

A unifying theme in my work is that robustness should not be treated as an afterthought. Instead, robustness must be embedded into the representation space, training objective, and evaluation protocol of the model. My Ph.D. research studies how compact quantum-enhanced representations, prototype-guided learning, uncertainty-aware weighting, and consistency-based objectives can help build AI systems that are more stable, interpretable, and secure under realistic clinical and computational constraints.

Current Research Program

My doctoral work has developed a coherent research program on **robust hybrid quantum-classical learning for medical image classification**. Rather than using quantum circuits as generic classifier replacements, I study their role as compact representation modules that can be integrated with classical deep neural networks. This perspective is especially relevant for near-term quantum machine learning, where practical progress depends on careful architectural design, stable training, and meaningful empirical evaluation.

My research contributions so far include three complementary directions:

- **Robust learning under noisy labels.** My MICCAI 2025 work, **NQNN**, studies multiclass medical image classification under symmetric label noise. The model integrates Fourier attenuation, uncertainty-based sample reweighting, and adaptive pooling into a variational quantum learning framework. This work addresses a common clinical problem: labels extracted from reports, expert annotations, or weak supervision pipelines may be inconsistent, incomplete, or noisy.
- **Robust learning under corrupted imaging inputs and weak supervision.** My AAAI 2026 work, **QAPNet**, extends this agenda from label corruption to image degradation. It combines sliding-window patch extraction, quantum patch encoding, additive-attention multiple-instance aggregation, contrastive alignment, and prototype regularization. This framework is designed to preserve class-discriminative structure under acquisition-time noise and patch-level uncertainty.
- **Secure learning under backdoor poisoning.** My current under-review work, **HQADNet**, studies training-time backdoor poisoning in medical image classification. It uses EM-refined soft prototypes, trust-aware learning, and an 8-qubit tree-structured quantum encoder to reduce trigger-driven shortcuts and encourage class-consistent representations. This direction connects robust medical AI with secure and adversarial machine learning.

In addition to model development, I have contributed to broader questions in hybrid quantum-classical learning. My *Journal of Imaging* work includes an empirical benchmark on barren plateau mitigation in quantum neural networks and a systematic review on systems-level support for hybrid quantum-classical learning with a medical imaging translation lens. These studies reflect my interest not only in model accuracy, but also in trainability, reproducibility, systems support, and deployable AI pipelines.

Research Themes and Future Directions

My future research agenda is organized around four connected themes.

1. Robust and secure AI for medical and biomedical data. I plan to continue developing models that can withstand label noise, input corruption, weak supervision, distribution shift, and adversarial manipulation. Future work will expand from benchmark medical images to larger clinical datasets, including chest radiographs, neuroimaging, and longitudinal biomedical records. A key objective is to design models that not only perform well on clean test data but also maintain reliability under realistic deployment stress.

2. Prototype-guided and uncertainty-aware representation learning. Across my work, prototypes serve as interpretable anchors for class structure, while uncertainty and trust scores help determine which samples should influence learning. I plan to extend this line of work toward calibrated prediction, open-set recognition, patient-level reliability analysis, and explainable medical AI. These methods can also support information visualization by making latent structure, class separation, and failure modes more transparent to clinicians and researchers.

3. Hybrid quantum-classical learning and high-performance AI systems. I am interested in the practical role of quantum components in modern AI pipelines. This includes variational quantum circuits, quantum kernels, quantum feature maps, quantum trainability, and simulation-efficient model design. My goal is to evaluate quantum modules carefully, compare them against strong classical baselines, and study when compact quantum representations provide useful inductive bias. This work naturally connects with high-performance computing for AI, reproducibility, and scalable experimentation.

4. Secure and trustworthy machine learning for safety-critical applications. My research increasingly connects robustness with security. Backdoor poisoning, adversarial perturbations, data leakage, and shortcut learning are important risks in medical AI and broader decision-support systems. I plan to study defense methods that combine representation learning, data-centric auditing, anomaly detection, and trust-aware training. This direction aligns with information security, secure computing, and responsible AI.

Broader Impact and Collaboration

My broader research goal is to make AI systems more reliable, interpretable, and useful in high-stakes domains. I am particularly motivated by problems where data are limited, noisy, imbalanced, or vulnerable to manipulation. Medical and biomedical AI offer an important testbed for these challenges, but the methods are broadly applicable to secure machine learning, software systems, intelligent data analysis, and trustworthy computing.

I also view research as a vehicle for student training. My prior academic appointments involved extensive undergraduate project and thesis supervision, and I plan to continue mentoring students through applied AI projects, reproducible experimentation, research writing, and interdisciplinary collaboration. I am especially interested in helping students connect foundational computer science concepts with modern research problems in AI, security, biomedical computing, and emerging computing systems.

Summary

My research program contributes to the development of trustworthy AI systems that are robust to imperfect data and security threats. Through work on noisy-label learning, corrupted inputs, weak supervision, backdoor poisoning, quantum trainability, and hybrid quantum-classical systems, I aim to build a foundation for reliable AI in medical and biomedical decision support. Going forward, I seek to advance research that combines methodological rigor, practical relevance, reproducibility, and student-centered collaboration.

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